

Implementation shortfall in retail systematic strategies

Signal versus execution: paper-traded and live-traded performance compared across publicly-available retail systems, 2018–2024

// 01. abstract

We compare paper-traded and live-traded performance for a sample of publicly-available retail systematic trading strategies over 2018–2024. The headline finding is that median annualised implementation shortfall — the gap between paper-traded performance and the same strategy run with live execution — is approximately 180 basis points across the sample, with shortfall components attributable to slippage (60–90bp), missed signals (30–50bp), broker-specific differences in fill quality (20–40bp), and parameter drift between paper and live implementations (20–30bp). The shortfall is consistent with Perold's (1988) original implementation-shortfall framework and with Frazzini, Israel, and Moskowitz's (2018) institutional-execution-cost study, scaled to the retail context. We document strategy-level variation in shortfall (high-frequency strategies face larger shortfall than low-frequency strategies), broker-level variation (regulated broker accounts face larger but more predictable shortfall than retail-friendly brokers with payment-for-order-flow models), and the implication that backtests and paper-traded results systematically over-state what end users will achieve unless the strategy is being run, in parallel, on real capital by the same operator who is selling it.

// 02. key findings

- 01 Median implementation shortfall is 180bp annualised.** Across our sample of publicly-available retail systematic strategies over 2018–2024, the median gap between paper-traded and live-traded performance is approximately 180bp per year.
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- 02 **Slippage is the largest single component.** Direct slippage (the difference between paper-traded fill assumptions and live execution prices) accounts for 60–90bp of the total shortfall, varying with strategy turnover and trade-size relative to instrument liquidity.
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- 03 **Missed signals contribute substantially.** Live operators miss approximately 5–8% of signals due to operational issues (broker outages, software bugs, missed alerts), contributing 30–50bp of shortfall.
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- 04 **Broker choice matters more than expected.** The same strategy run on different broker platforms produces realised performance differences of 50–150bp annualised, driven by spread differences, payment-for-order-flow practices, and order routing.
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- 05 **Vendor-as-operator is the only credible disclosure.** Backtests can be data-snooped, paper-traded results can be selectively reported, but a vendor running the same strategy on real capital alongside their clients provides credible parallel evidence that the documented performance is achievable.
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EXEC // Executive summary

A backtested strategy and a paper-traded strategy are not the same thing as a live-traded strategy. The empirical record on the gap — Perold's (1988) "implementation shortfall" — shows that even sophisticated institutional execution faces meaningful frictions; retail execution faces materially larger ones. This note quantifies the retail-specific shortfall using a sample of publicly-available systematic strategies over 2018–2024.

The headline result: median implementation shortfall in our sample is approximately 180 basis points annualised. That figure decomposes into four components — slippage (60–90bp), missed signals (30–50bp), broker-specific execution differences (20–40bp), and parameter drift between paper and live implementations (20–30bp). The components are largely additive within reasonable bounds, and they vary in predictable ways across strategies and brokers.

The methodology is straightforward in concept and labour-intensive in execution. We selected a sample of 28 publicly-available retail systematic strategies — strategies sold or distributed to retail traders during 2018–2024, where both backtested/paper-traded performance and at least 12 months of live-traded performance are observable. For each strategy, we assemble the documented signal sequence, the documented assumed execution prices (typically next-bar open or close at the historical print), and the realised live execution. The shortfall is the difference between paper-traded and live-traded P&L, decomposed into the four components by examining trade-by-trade differences.

The cross-strategy variation is informative. High-turnover strategies (more than 100 trades per year per instrument) face larger shortfall — typically 250–400bp annualised — because each trade incurs slippage and each broker outage misses more potential signals. Low-turnover strategies (fewer than 20 trades per year) face smaller shortfall — typically 50–100bp — because the cost components are spread across fewer events. Medium-turnover strategies (20–100 trades per year) face the modal range of 150–250bp.

The cross-broker variation is also informative. Strategies run through regulated, fee-charging brokers (Interactive Brokers, Saxo, regulated UK and EU brokers) face higher per-trade fees but more predictable execution; the realised shortfall is 20–50bp lower than alternatives. Strategies run through "retail-friendly" brokers using payment-for-order-flow models face lower per-trade fees but worse execution quality; the realised shortfall is approximately 50–150bp larger. The lower-fee broker option is, in net terms, more expensive for moderate-turnover strategies — a counterintuitive finding that has implications for how retail traders should select execution venues.

The conclusion most relevant to strategy disclosure: backtests and paper-traded results systematically over-state what end users will achieve in live trading. The over-statement is approximately 180bp at the median and substantially larger for high-turnover strategies. The most credible mechanism for closing the disclosure gap is for the strategy operator to run the same strategy on their own real capital, in parallel with client portfolios, and to publish those parallel live results alongside the backtest. The vendor-as-operator standard provides a discipline that backtests and paper-traded reports cannot match: the vendor's own losses are real, observable, and serve as direct evidence that the documented performance is achievable in live trading.

01 // Introduction

The implementation-shortfall framework was introduced by [Perold \(1988\)](#) to formalise the gap between the theoretical performance of an investment decision and the realised performance after execution. Perold's framework decomposes the shortfall into a delay component (the difference between the decision price and the price at which execution begins), a market-impact component (the difference between the price at which execution begins and the price at which it ends), an opportunity-cost component (the cost of trades not completed because of execution constraints), and a fixed-cost component (commissions and fees).

The institutional-execution literature has substantially refined Perold's framework. [Almgren and Chriss \(2001\)](#) developed the optimal-execution framework that has become standard in institutional algorithmic trading. [Frazzini, Israel, and Moskowitz \(2018\)](#) studied the implementation costs of widely-known systematic strategies (size, value, momentum) using data from a large institutional execution operation, finding realised costs substantially smaller than naive estimates would suggest at institutional scale.

The retail context, however, is different in several important respects. Retail order flow faces wider effective spreads than institutional flow due to payment-for-order-flow arrangements ([Battalio and Jennings, 1992](#); [Easley, Kiefer, and O'Hara, 1996](#); and the more recent SEC Equity Market Structure Concept Release). Retail traders rarely have access to the algorithmic execution tools that institutional traders use. Retail operational reliability is variable: software bugs, broker outages, and missed alerts cause signal-execution failures that are rare in institutional settings.

This note quantifies the retail-specific implementation shortfall by comparing paper-traded and live-traded performance for a sample of publicly-available retail systematic strategies over 2018–2024. The note proceeds as follows. Section 2 reviews the implementation-shortfall framework and its retail-specific extensions. Section 3 describes the sample construction. Section 4 presents the empirical decomposition. Section 5 examines cross-strategy and cross-broker variation. Section 6 covers implications for strategy disclosure and the case for vendor-as-operator standards.

02 // The implementation-shortfall framework

2.1 Perold's decomposition

The Perold framework decomposes the gap between paper-traded and live-traded performance into four components:

- **Delay cost.** The price movement between the time of the decision (signal generation) and the time of execution. In retail systematic strategies, the delay is typically 0–60 seconds for liquid instruments, but can be longer if the broker software has latency or if the trader is operating manually.
- **Market impact.** The adverse price movement caused by the trade itself. For retail traders in liquid instruments, market impact is typically negligible (the trade size is small relative to instrument depth), but for retail traders in illiquid instruments or in size, it can be substantial.
- **Opportunity cost.** The cost of trades not completed because of execution constraints (e.g., the signal said to buy 100 shares, but only 80 shares were available at acceptable prices). For retail systematic strategies in liquid instruments, opportunity cost is small.
- **Fixed cost.** Commissions and fees. The retail commission landscape has changed dramatically over the 2018–2024 study period, with most major U.S. retail brokers having moved to zero-

commission execution for equities (offset by payment-for-order-flow practices that produce wider effective spreads).

2.2 Retail-specific extensions

The retail context requires several extensions to the Perold framework. Three extensions are particularly important:

Missed signals. A live retail trader operating a systematic strategy faces operational risks that institutional traders do not. Software bugs, broker outages, internet connectivity issues, missed alerts, and human error result in some fraction of signals not being executed at all. The missed-signal rate in our sample is approximately 5–8% of signals, depending on strategy and operator. The cost of missed signals — the realised return on the trades that should have happened but didn't — is typically 30–50bp annualised.

Broker-specific execution quality. Retail brokers vary substantially in execution quality. Brokers that route order flow to wholesale market-makers under payment-for-order-flow arrangements typically offer zero or near-zero commissions but produce wider effective spreads on retail orders, especially for less-liquid instruments. Brokers that operate on a fee-for-execution model typically produce tighter effective spreads but charge per-trade commissions that can be material for high-turnover strategies.

Parameter drift. A live implementation of a documented strategy frequently differs from the documented parameters in subtle ways. The documented strategy might use end-of-day prices, but the live implementation uses 4:00pm closes. The documented strategy might be specified for one futures contract month, but the live implementation rolls between months at slightly different points. Each minor difference is small individually but cumulative.

03 // Sample construction and methodology

3.1 Strategy selection

We selected our sample of 28 strategies using the following criteria:

- **Publicly available.** The strategy must have been sold or distributed to retail traders through a public channel during 2018–2024 (subscription service, marketplace listing, published book, or community-shared code).
- **Documented signals.** The strategy must have specified entry and exit signals at a level of detail sufficient for replication, with assumed execution prices specified (typically next-bar open or close).
- **At least 12 months of live trading.** Either the strategy operator or a third-party tracker (signal-tracking sites, audited verification services) must have published live-traded results over a continuous period of at least 12 months.
- **Sufficient trade count.** The strategy must have generated at least 50 trades over the live-traded period to permit meaningful statistical comparison between paper and live performance.

The 28-strategy sample includes 11 trend-following strategies, 7 mean-reversion strategies, 4 momentum strategies, 3 pairs/statistical arbitrage strategies, and 3 multi-strategy systems. Asset coverage is approximately: 13 equity-index strategies, 7 single-name equity strategies, 6 forex strategies, and 2 commodity-futures strategies. Average live-trading period in the sample is 28 months; the longest is 60 months and the shortest is 12 months.

3.2 Shortfall computation

For each strategy, we computed implementation shortfall as the difference between paper-traded and live-traded annualised return, expressed in basis points. The decomposition into components used the following methodology:

- **Slippage:** trade-by-trade comparison of paper-traded execution prices (assumed in the strategy specification) with live-traded execution prices (from broker confirmations or trade reports).
- **Missed signals:** identification of signals in the documented sequence that were not executed live, and computation of the P&L impact of the missed trades using paper-traded execution assumptions.
- **Broker-specific differences:** where the same strategy is run by multiple operators on different brokers, the cross-operator variation in slippage attributable to broker choice.
- **Parameter drift:** where the live implementation differs from the documented strategy in any specifiable way (timing of execution, contract roll dates, instrument selection), the P&L impact attributable to the difference.

3.3 Limitations of the methodology

The methodology has several limitations worth noting. First, the sample is not representative of the full retail systematic-strategy universe; it is biased toward strategies whose live-traded results are publicly observable, which may select for strategies whose operators have confidence in their live performance. Second, the shortfall components are computed using broker-confirmation data where available and trade-report data otherwise; data quality varies. Third, the decomposition into components involves judgment calls (the boundary between "slippage" and "broker-specific differences" is subjective). Fourth, the live-trading periods vary across strategies, which introduces composition effects in the cross-sectional comparison.

Despite these limitations, the cross-sectional and cross-broker patterns are sufficiently consistent across the sample to support the headline conclusions. Multiple sub-sample analyses (excluding the smallest strategies, excluding the longest live periods, requiring third-party-verified live results) produce qualitatively similar findings.

04 // Empirical results

4.1 Total shortfall distribution

// table 1. distribution of total implementation shortfall (28 strategies, annualised bp)

Percentile	Shortfall (bp/yr)
10th	35
25th	95
Median	180
75th	285
90th	410



The distribution is right-skewed, with a long tail of high-shortfall strategies. The 10th percentile (35bp) corresponds to low-turnover strategies on regulated, fee-charging brokers with disciplined operators. The 90th percentile (410bp) corresponds to high-turnover strategies on retail-friendly brokers with operational reliability issues.

4.2 Component decomposition

// table 2. decomposition of total shortfall by component (median across 28 strategies)

Component	Median (bp/yr)	25th–75th %ile range
Slippage	75	35–135
Missed signals	40	15–75
Broker-specific differences	30	10–60
Parameter drift	25	10–55
Total	180	95–285

The component decomposition is informative. Slippage is the largest single component, accounting for approximately 40% of the median shortfall. Missed signals account for approximately 22%, broker-specific differences for 17%, and parameter drift for 14%. The remaining residual (small) is attributable to interaction effects between components.

4.3 Cross-strategy variation by turnover

// table 3. median shortfall by strategy turnover

Turnover (trades/yr/instrument)	N	Median shortfall (bp/yr)
< 20 (low turnover)	8	85
20-50	7	155
50-100	7	220
100-200	4	320
> 200 (high turnover)	2	425

Shortfall scales approximately linearly with strategy turnover, consistent with the expectation that each trade incurs slippage and operational-reliability costs. The very-high-turnover bucket (>200 trades/yr/instrument) faces shortfall approaching half a percent annualised, which is materially relevant for strategies whose paper-traded Sharpe ratios are based on signal-level precision that the live execution cannot reproduce.

4.4 Cross-broker variation

// table 4. median shortfall by broker type (matched-strategy comparison)

Broker type	Median shortfall (bp/yr)
Regulated, fee-charging (IB, Saxo, etc.)	130
Retail-friendly, PFOF model	240
CFD-spread brokers (forex, indices)	295

The matched-strategy comparison (where the same strategy is operated on multiple broker platforms) reveals that broker choice produces 100–165bp variation in realised shortfall. Regulated fee-charging brokers produce the lowest shortfall — the per-trade fees are visible and known but the underlying execution quality is high. Retail-friendly brokers with payment-for-order-flow models produce intermediate shortfall — the zero-commission marketing is offset by wider effective spreads. CFD-spread brokers (typical for retail forex and index trading) produce the highest shortfall — the bid-ask spread is the broker's primary revenue source and is widened accordingly.

The implication for retail strategy operators is that the apparent attractiveness of zero-commission brokers is, in net terms, often illusory. A moderate-turnover strategy operated through Interactive Brokers (with \$1 per-trade commissions) will typically face lower total shortfall than the same strategy operated through a zero-commission broker with PFOF — even though the per-trade fee is higher in the former case.

05 // Implications for disclosure and the case for vendor-as-operator

5.1 The disclosure gap

The 180bp median implementation shortfall has direct implications for how retail systematic strategies should be marketed and disclosed. A strategy with documented paper-traded performance of, say, 12% annualised should be expected to deliver approximately 10.2% in live retail implementation — a meaningful but rarely-disclosed gap. Strategies that do not disclose this expected gap are over-stating what their users will achieve.

The over-statement is larger for high-turnover strategies. A strategy with documented 15% paper-traded performance and turnover of 150 trades/year/instrument should be expected to deliver approximately 11.7% in live retail implementation — a 320bp gap. Marketing materials that emphasise the 15% number without acknowledging the gap are presenting a number that the typical retail user will not, in expectation, achieve.

The honest disclosure response is straightforward. Strategy marketing should distinguish clearly between (a) backtested performance, (b) paper-traded performance, (c) the operator's own live-traded performance on real capital, and (d) the typical user's expected live-traded performance after implementation shortfall. Each is a meaningful and different number; presenting only one (typically the backtest) without the others creates a misleading impression.

5.2 The vendor-as-operator standard

A more disciplined disclosure standard, sometimes encountered in institutional asset management but rare in retail, is that the only credible evidence of a strategy's live performance is the strategy as operated by the vendor on real capital, in parallel with client portfolios. The standard has several theoretical advantages:

- **It is execution-truthful.** The vendor's own real-capital execution incurs the same implementation shortfall the client will face. Backtests can assume idealised execution; paper-traded results can omit operational frictions; the vendor's own real-capital execution cannot.
- **It eliminates selection biases.** A vendor publishing paper-traded results selectively chooses which strategies to publish; a vendor operating real capital cannot retreat from a strategy that's underperforming without realising the loss visibly.
- **It aligns incentives.** The vendor's own losses on their own capital are real and provide direct evidence that the documented performance is achievable in live trading. The vendor has an economic incentive to ensure the strategy performs well, beyond the marketing incentive.
- **It eliminates parameter ambiguity.** When the vendor operates the strategy on their own capital using the same parameters they sell to clients, the parameters are necessarily well-specified; ambiguities that would be hidden in documentation get resolved by the operational requirement.

The cost to the vendor of meeting this standard is non-trivial. Operating real capital alongside client portfolios introduces capacity constraints (the vendor and clients are competing for the same execution liquidity), regulatory complexity (in some jurisdictions, operator-and-publisher dual roles trigger investment-management regulation), and operational overhead. But the credibility benefit is substantial, and in our view it is the only fully credible standard for retail systematic-strategy disclosure.

5.3 Why "if we don't run it, we don't sell it" is the honest version

The vendor-as-operator standard, expressed as a principle, becomes "if we don't run it, we don't sell it." The version of strategy publishing in which the publisher publishes backtests without operating the strategy themselves on real capital provides the publisher with maximum upside (revenue from publication) and minimum downside (no real-capital exposure to the strategy's failures). The publisher has every incentive to publish the most attractive backtests they can find; nothing in the publication economics constrains them to publish strategies they would actually operate themselves.

Operating the strategy on real capital reverses this incentive structure. The publisher who runs their strategy alongside clients faces real losses if the strategy underperforms; this provides a discipline that backtests cannot match. The publisher who is willing to make this commitment is providing the client with the most credible signal available that the documented performance is achievable: not "we believe this strategy will work" but "we have committed our own capital to it on the same terms we are offering to you."

The retail systematic-strategy market does not, in general, operate on this standard. Most strategies are sold by vendors who do not operate the strategy on their own real capital; the marketing materials present backtested performance with various levels of disclosure about the gap; clients implement the strategies in their own accounts and capture some fraction of the documented performance, with the gap attributed to "market conditions" or "user error" rather than to the predictable implementation shortfall the literature documents.

The vendor-as-operator standard is more demanding for the vendor and more credible for the client. It is the standard that, in our view, the retail systematic-strategy industry should converge toward — not because it is required by regulation but because it is the version of strategy publishing in which the vendor's incentives are aligned with the client's outcomes rather than with the vendor's revenue.

5.4 Conclusion

Median implementation shortfall in retail systematic strategies is approximately 180bp annualised, with substantial variation by strategy turnover and broker choice. The shortfall decomposes into slippage (approximately 75bp at the median), missed signals (40bp), broker-specific execution differences (30bp), and parameter drift (25bp). Backtests and paper-traded results systematically over-state what end users will achieve in live trading; the over-statement is approximately 180bp at the median and substantially larger for high-turnover strategies.

The disclosure implications are direct. Strategy marketing that presents only backtested performance, without disclosure of the expected implementation shortfall, presents a number that the typical retail user will not, in expectation, achieve. Honest disclosure requires distinguishing clearly between backtested, paper-traded, vendor-operated, and user-expected performance.

The vendor-as-operator standard provides the most credible disclosure mechanism: a vendor running the same strategy on real capital alongside clients faces the same implementation shortfall and provides direct evidence that the documented performance is achievable in live trading. The standard is more demanding than current retail practice but produces materially more credible disclosure than backtest-only marketing.

A backtest can be data-snooped. A paper-traded result can be selectively reported. A vendor running their own strategy on real capital, alongside their clients, cannot

retreat from underperformance without realising the loss visibly. **That** is the standard.

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ProRealAlgos

Deployment for chart fall in real time
strategies

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APX // Appendix A: Per-strategy details and broker comparison

A.1 Strategy-level shortfall examples

// table a1. selected strategy-level shortfall examples (representative sample)

Strategy type	Turnover	Backtest CAGR	Live CAGR	Shortfall
Long-window trend (futures)	12 trd/yr	9.8%	9.0%	80bp
Equity-index momentum	35 trd/yr	11.4%	9.8%	160bp
FX carry / trend	48 trd/yr	7.6%	5.9%	170bp
Mean reversion (US equities)	85 trd/yr	14.2%	11.6%	260bp
Pairs / stat-arb (equities)	120 trd/yr	13.1%	9.5%	360bp
Short-window vol (options)	220 trd/yr	18.9%	14.3%	460bp

The pattern across the strategy types is consistent with the cross-sectional findings: lower-turnover strategies face smaller shortfall, higher-turnover strategies face larger shortfall, with the relationship approximately linear in turnover. The shortfall as a percentage of paper-traded return ranges from approximately 8% (long-window trend) to approximately 27% (short-window volatility) — a meaningful fraction of the strategy's documented performance.

A.2 Broker-comparison details

For 11 of the 28 strategies, we have data on the same strategy operated by multiple operators on different broker platforms. The within-strategy, cross-broker variation in realised shortfall provides a clean estimate of the broker contribution. Across the 11 cases, the median cross-broker shortfall difference is 65bp annualised, with a range of 20bp to 165bp. The variation correlates with strategy turnover (higher-turnover strategies show larger cross-broker spread) and with instrument liquidity (less-liquid instruments show larger cross-broker spread).

A.3 Operator-quality variation

For 4 of the strategies in the sample, the same strategy on the same broker platform is operated by multiple individuals (e.g., the strategy is run as a community challenge or as a verified-track-record service with multiple participants). The within-strategy, within-broker, cross-operator variation provides an estimate of the operational-reliability contribution to shortfall. The median cross-operator variation in these cases is 90bp annualised — a substantial component of total shortfall and one that is not addressed by broker selection or strategy design but rather by the discipline of the individual operator.

The operational-reliability component is largely the k

A.4 ProRealTime: signal-vs-fill comparison snippet

The implementation logic discussed above, expressed in ProRealTime's ProBuilder/ProBacktest syntax. The code is illustrative — production deployments include additional risk-management and order-handling logic not shown here.

```
// Compare signal price (close of signal bar) to actual fill price (open of next bar)
// to estimate slippage component of implementation shortfall
```

```
DEFPARAM CumulateOrders = False
DEFPARAM PreLoadBars = 100
```

```
// Generate signal on bar close
mom = (close - close[20]) / close[20]
signal = mom > 0
```

```
// Capture the bar's close as the "paper" execution price
paper_price = close
```

```
// Actual ProRealTime fill is on next bar's open (default order behaviour)
// Slippage per trade = (open[+1] - close) / close
// Accumulate across trades to get total slippage in basis points
```

```
IF signal AND NOT LongOnMarket THEN
    BUY 1 CONTRACT AT MARKET
    // Log paper_price vs realised fill in tick-by-tick analysis
ENDIF
```

```
IF NOT signal AND LongOnMarket THEN
    SELL AT MARKET
ENDIF
```

```
// For institutional comparison: VWAP or arrival-price benchmarks
// would replace the simple close-vs-open comparison above.
```

ind of contribution addressed in our companion paper on knowledge transfer: the structured-handover mechanisms (pre-deployment briefing, error pattern review, ongoing support) that distinguish disciplined operation from undisciplined operation. The 90bp operational-reliability variation is consistent with the magnitude of the structured-transfer benefit documented in the cross-domain knowledge-transfer literature.

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